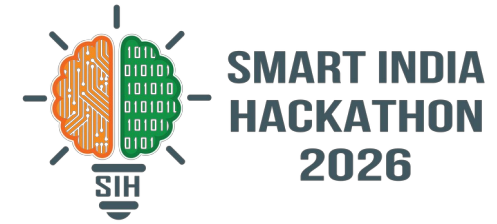


SMART INDIA HACKATHON 2026



- **Problem Statement ID** – SIH26138
- **Problem Statement Title** – Quantum-inspired fuel consumption prediction and green fleet optimization
- **Theme** – Clean and Green Technology
- **PS Category** – Software
- **Team ID** – S16
- **Team Name** – CloudBusters

TEAM MEMBERS

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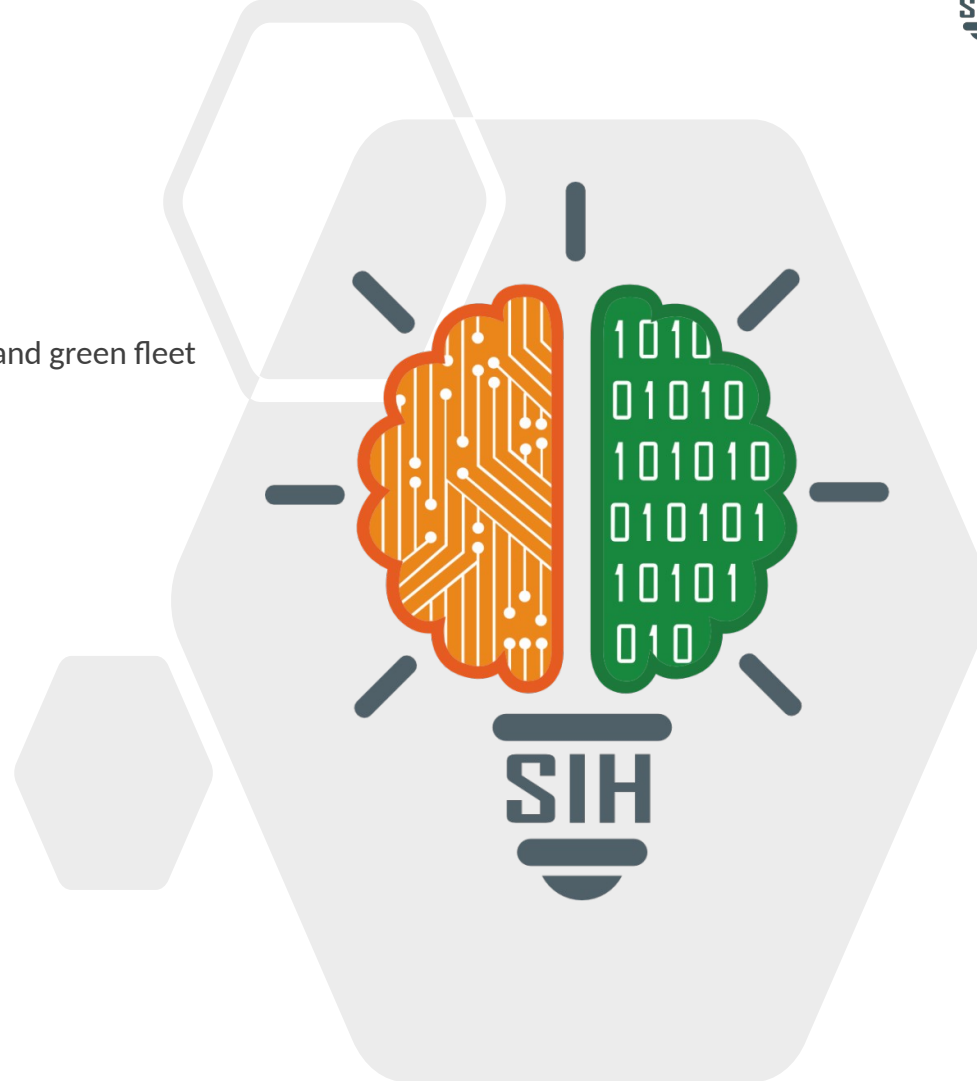
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Anshika Singh 2415000265

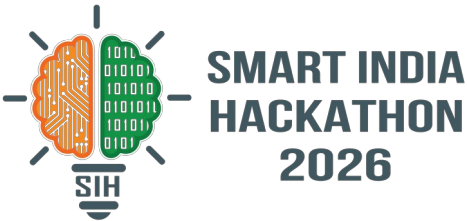
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QFLEET



❖ Proposed Solution (Describe your Idea/Solution/Prototype)

DETAILED EXPLANATION — ONE OPTIMIZATION LOOP

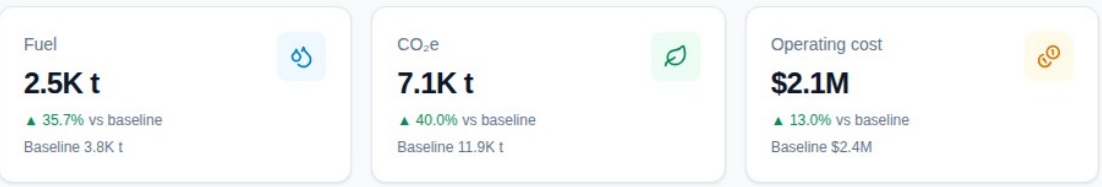
- 1. **ASSIGN** — QIEA assigns vessels, routes and fuels — the discrete decisions.
- 2. **TUNE** — QPSO tunes sailing speed (continuous) in the same loop.
- 3. **EVOLVE** — Q-NSGA-II evolves a full Pareto front across cost, well-to-wake GHG and schedule reliability.
- 4. **COMPLY** — Every Pareto point ships with its CII rating, EEXI pass/fail and FuelEU gap computed.

GROUNDING IN REAL PHYSICS — NOT A BLACK BOX

- Admiralty V^3 law and resistance modelling — hull fouling, draft and sea state.
- Full tank-to-wake → well-to-wake accounting for every candidate solution.

HOW IT ADDRESSES THE PROBLEM

- Cost, GHG and schedule reliability are optimized jointly, not traded off ad hoc.
- Operators pick a point on the Pareto front instead of accepting one fixed answer.
- Compliance (CII / EEXI / FuelEU) is computed inside the loop, not bolted on after.



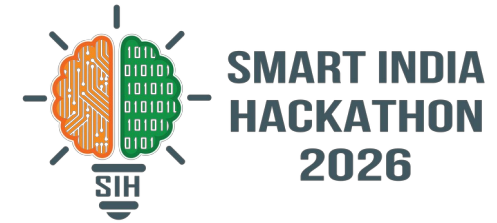
Prototype output — fuel, CO₂e and cost against the greedy baseline

INNOVATION AND UNIQUENESS

- MILP cannot represent the non-convex V^3 speed-fuel curve; QFleet can.
- ML baselines only forecast — QFleet decides speed, fuel and deployment.
- Single-objective GA / PSO bolt emissions on afterwards — QFleet is Pareto-native.
- QUBO formulation is hardware-ready: a real annealer is a backend swap.



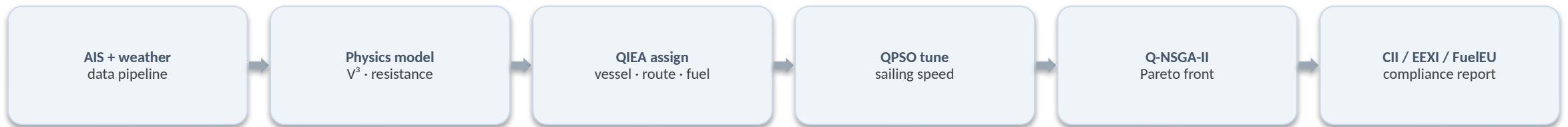
TECHNICAL APPROACH



TECHNOLOGIES TO BE USED

Optimization Core	pymoo • pyqubo • dwave-neal	Hardware-ready QUBO; simulated annealing keeps demo runtime deterministic
Quantum-Inspired ML	PennyLane • Qiskit-Aer	Small statevector circuits (≤ 10 qubits) as feature maps — fast to simulate
Classical ML Baseline	XGBoost • LightGBM • SHAP	The benchmark QFleet must beat, with SHAP plots for interpretability
Data Pipeline	Polars • GeoPandas • Xarray	AIS + Copernicus weather at scale, stored in Parquet — not CSV
Backend	FastAPI • Celery • Redis	Optimization runs (2–5 min) go to an async job queue, never a blocking request
Reporting & Infra	WeasyPrint • Docker Compose	Auto-generated CII / EEXI / FuelEU compliance PDFs; one command boots the stack

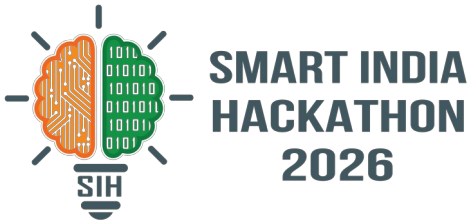
METHODOLOGY AND PROCESS FOR IMPLEMENTATION



Runs entirely on a laptop — quantum-ready, classically executed



FEASIBILITY AND VIABILITY



FEASIBILITY OF THE IDEA

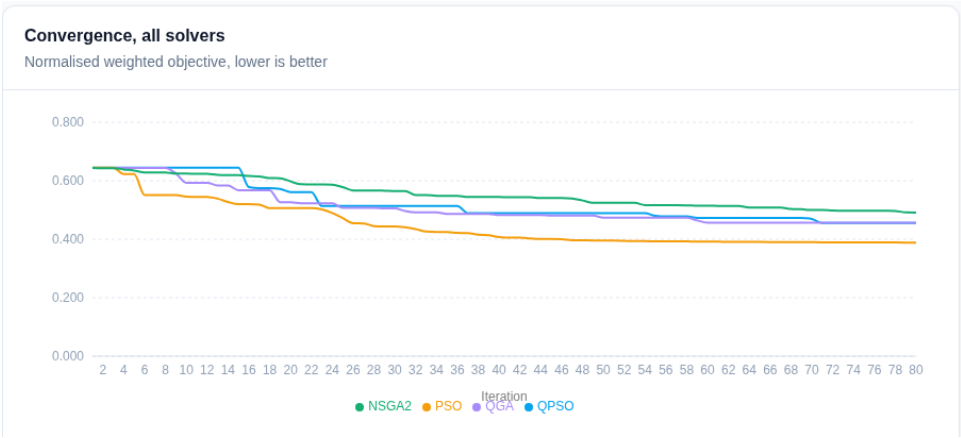
- Contract-first architecture — one shared schema
- Async job queue absorbs 2–5 min runs
- Runs entirely on a laptop
- Deterministic, reproducible demo runs
- Physics-calibrated data vs MRV aggregates

POTENTIAL CHALLENGES AND RISKS

- Classical annealing only — no real annealer yet
- LNG + methanol prioritised over hydrogen
- Fewer benchmark seeds than a full study
- Synthetic data unvalidated at fleet scale
- Multi-objective runtime grows with fleet size

STRATEGIES TO OVERCOME THEM

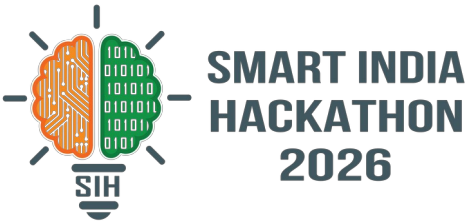
- QUBO is a backend swap — annealers plug in
- Pre-compute demos; optimize small sets live
- Extend fuels to hydrogen and ammonia next
- Live AIS + weather for continuous re-runs
- Multi-seed benchmarks with error bars



Solver convergence on one problem instance — measured in the working prototype



IMPACT AND BENEFITS



IMPACT ON THE TARGET AUDIENCE

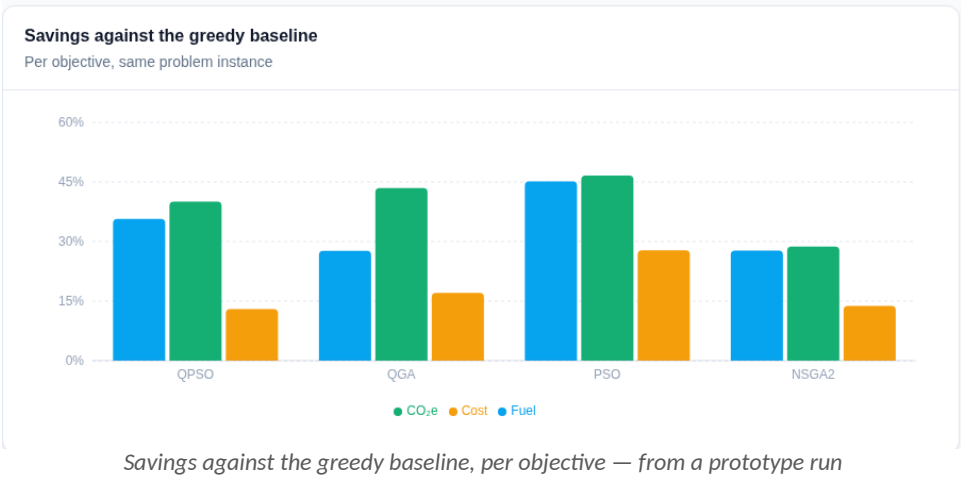
- Operators — lower fuel cost per voyage
- Compliance teams — CII / EEXI evidence
- Charterers — cleaner, priced-in voyages
- Port planners — reliable, lower-emission calls
- Regulators — auditable decisions

BENEFITS OF THE SOLUTION

- Well-to-wake accounting incl. CH₄ and N₂O
- Multi-fuel: VLSFO / MGO / LNG / methanol
- Avoids EU ETS and FuelEU penalty exposure
- Cleaner coastal air near ECAs
- Transparent, audit-ready recommendations

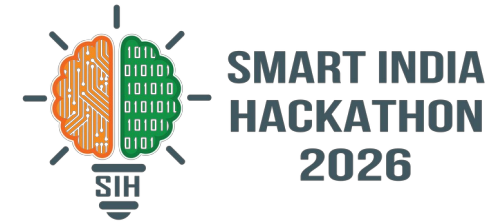
HOW WE MEASURE IMPACT

- Fuel cost per voyage
- Well-to-wake CO₂e per route
- CII rating spread across the fleet
- Schedule reliability (%)
- EU ETS + FuelEU penalty exposure





RESEARCH AND REFERENCES



REGULATION AND STANDARDS

- IMO MEPC — Carbon Intensity Indicator (CII) rating framework.
- IMO — Energy Efficiency Existing Ship Index (EEXI).
- EU FuelEU Maritime — GHG intensity limits for marine fuels.
- EU Emissions Trading System (ETS) extension to shipping.
- MARPOL Annex VI — sulphur and emission control areas.

ALGORITHMS AND METHODS

- Sun, Feng & Xu (2004) — quantum-behaved particle swarm optimization (QPSO).
- Han & Kim (2002) — quantum-inspired evolutionary algorithm (QIEA).
- Deb et al. (2002) — NSGA-II, IEEE Trans. Evolutionary Computation.
- Admiralty coefficient / V^3 speed–power law for propulsion resistance.
- QUBO formulation for quantum annealing.

DATA AND TOOLING

- AIS vessel movement data for routes and speed profiles.
- Copernicus Marine / ERA5 weather and sea-state reanalysis.
- IMO MRV and EU THETIS-MRV aggregate emissions reporting.
- pymoo, pyqubo, dwave-neal — optimization and annealing libraries.
- PennyLane, Qiskit-Aer — quantum circuit simulation.